

ARMAAN GUPTA

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Education

University of Waterloo

Bachelor of Computer Science + Minor in Artificial Intelligence

Sept. 2024 – Apr. 2029

Waterloo, ON

Technical Skills

Languages: Java, Python, C, C++, C#, SQL, JavaScript, TypeScript, HTML/CSS, Swift, PHP, Ruby on Rails

Frameworks, Tools, & Libraries: React, Node.js, Git, AWS, TensorFlow, PyTorch, Firebase, Angular, Docker Next.js, npm, Flask, pandas, NumPy, MongoDB, OpenCV, Tailwind, FastAPI, REST API, Vite, Three.js, Figma, Linux, Bash, GCP

Experience

Ford

Jan. 2026 – Present

Software Engineer

Toronto, ON

- Eliminated manual order ingestion paths by designing automated validation and **persistence pipelines**, reducing human intervention by **60%+**.
- Instrumented distributed services with Datadog APM (tracing, metrics, alerts), enabling **low-latency** debugging and proactive performance **regression detection**.

Baba

Sept. 2025 – Jan. 2026

Engineer

New York, NY

- Joined as one of the **first engineers**, building core infrastructure that facilitated a **\$4M+ Pre-Seed**
- Engineered the most comprehensive **care-center search platform** in the USA, aggregating data across **40,000+** facilities, generating **10,000+** visits within weeks.
- Engineered a prescription **intelligence engine** using SQLAlchemy, and Supabase that aggregates free, reliable public APIs to identify generic alternatives, ensuring data accuracy while **eliminating licensing costs**.

WATonomous

July 2025 – Present

Software Engineer

Waterloo, ON

- Built an Automated Power Outage Notification System, leveraging **cloud-based services** and containerized microservices with Docker, reducing manual reporting delays by **80%**.
- Developed cloud-native APIs and automation scripts to integrate with SLURM and Kubernetes, enabling **parallel execution** of simulation jobs and increasing throughput by **300%**.

Government of Canada

May 2025 – Aug. 2025

Software Engineer

Toronto, ON

- Developed a climate reanalysis model pipeline by integrating **ML-based** interpolation techniques using **scikit-learn** and **PyTorch**, improving **spatial resolution** of missing environmental data by **35%**.
- Built an interactive web tool to explore **40+ years** of climate data across **50+** stations and **16 datasets**.

Projects

🏆 **arXivisual** | FastAPI, Modal, SQLAlchemy, TypeScript, Multi-agent Pipelines

🔗 **repo link**

- Built a full-stack AI platform that transforms **arXiv research papers** into 3Blue1Brown-style animated explanations using a **7-agent pipeline** with Modal serverless video rendering.
- Engineered a **4-stage quality validation** system (AST parsing, spatial bounds, LLM narration scoring, render testing), achieving **98%** visualization success and **100,000+** views with **10,000+** users in 1 week.

Assembl3D | Next.js, Three.js, Gemini Vision, Node.js, Bright Data

🔗 **repo link**

- Built a tool to convert 2D PDF manuals into interactive 3D visualizations, leveraging a **microservices architecture** with Next.js and **React Three Fiber**.
- Engineered a scraping pipeline to harvest product manuals, feeding a multi-modal Gemini vision system to extract **3D spatial geometry** and assembly logic.